

Purchase-aware Tree Distillation for Home-feed Retrieval

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ABSTRACT

Large-scale home feeds commonly separate candidate retrieval from a purchase-oriented ranker. This division creates an objective gap: a retriever is optimized from sparse interactions while the downstream ranker estimates a denser purchase utility. We formulate Purchase-aware Tree Distillation (PTD), a preregistered tree-retrieval method that transfers a frozen Entire Space Multi-Task Model (ESMM) teacher into local sibling decisions of a balanced item tree. PTD combines item-sibling and internal-node-sibling distillation, an HSTU-style action-sequence encoder, and a train-only alternating tree update related to TDM/JTM. The evaluation contract fixes a point-in-time-safe evidence universe, a five-date temporal test, three random seeds, paired user bootstrap inference, and explicit quality, diversity, and latency guardrails. A verified legacy artifact establishes the common evidence universe and teacher anchor (12,844,800 test pairs; ESMM purchase NDCG@50 0.17626), but it contains no PTD result. The prospective five-day tree-family results are pending; consequently this draft makes no efficacy or novelty claim beyond the registered method and its distinctions from prior work. The accompanying artifact provides schemas, a claim-evidence ledger, deterministic table/figure generation, and fail-closed validation so that empirical claims can be promoted only after immutable manifests, hashes, and evaluation JSON are available.

CCS CONCEPTS

• **Information systems** → **Recommender systems**; **Retrieval models and ranking**; • **Computing methodologies** → *Knowledge representation and reasoning*.

KEYWORDS

candidate retrieval, recommender systems, tree-based retrieval, knowledge distillation, conversion modeling, sequential recommendation

1 INTRODUCTION

Industrial recommendation pipelines must retrieve a small candidate set before an expressive ranker can evaluate it. Two-tower retrieval makes this stage compatible with maximum-inner-product indexes, but constrains the interaction function. Tree-based deep models instead place items at leaves and evaluate user-node compatibility from root to leaf, supporting expressive scorers with logarithmic dependence on corpus size under fixed branching and beam assumptions [9]. Their supervision, however, is usually derived from observed positive paths and sampled negatives. The downstream purchase ranker may encode a richer ordering than those binary path labels.

This paper studies a deliberately narrow question: can a fixed purchase model teach the local decisions made by a tree retriever? We call the registered method Purchase-aware Tree Distillation (PTD). The teacher is a frozen ESMM-style model whose purchase score factors as predicted click-through rate times predicted post-click conversion rate [3]. The student learns distributions over item siblings and internal-node siblings. Local normalization matches

the decision geometry of top-down retrieval and avoids requiring the teacher to define a globally normalized distribution over the complete catalog.

Three components are integrated. First, an HSTU-style encoder represents the user’s point-in-time action sequence [7]. Second, a balanced tree constrains serving work and makes sibling groups explicit. Third, an optional JTM-style alternating phase updates item assignments from training evidence only and locks the tree before test scoring [8]. Figure 1 summarizes the design.

This is an evidence-bounded draft. We contribute (i) a precise method and training algorithm, (ii) a preregistered temporal evaluation and ablation plan, and (iii) a reproducible artifact contract that separates verified, preregistered, and pending statements. We do *not* claim that the combination is the first of its kind or that it improves retrieval: those judgments require the pending prospective evaluation and a broader novelty audit.

2 PROBLEM FORMULATION

Let u denote a request with a history H_u available strictly before snapshot time t , and let I_t be eligible items. A balanced b -ary tree $\mathcal{T}$ maps each item i to a leaf. At depth h , beam search keeps B nodes according to a student score $s_\theta(u, n)$. The serving objective is to retrieve a set $R_K(u)$ with high future purchase utility while bounding the number of node evaluations.

Write the root-to-leaf code for item i as $\pi_{\mathcal{T}}(i) = (n_0, \dots, n_H)$ and the children competing with n_h as $S_h(i)$. With a balanced tree, $H = \lceil \log_b |I_t| \rceil$ up to capacity padding. After encoding the user once, beam retrieval evaluates at most $b + Bb(H - 1)$ child nodes before leaf truncation, or $O(BbH)$ scorer calls. This bound motivates a local decision objective: the student need only order siblings encountered by the beam, rather than score every eligible item online.

For each eligible pair, the frozen teacher produces

$$q_T^{\text{purchase}}(u, i) = \widehat{p}_T(\text{click} \mid u, i) \widehat{p}_T(\text{purchase} \mid \text{click}, u, i), \quad (1)$$

following the ESMM factorization. This score is a training signal, not ground truth. It can inherit calibration, exposure, and representation biases from the legacy model.

The observed label y_{ui} is one only when a home-feed click-to-purchase event occurs in $[t, t + 24\text{h})$. All sequence, item, teacher-feature, and tree-construction inputs must precede t . Our primary metric is macro user purchase NDCG@50 over users with an observed purchase.

3 PURCHASE-AWARE TREE DISTILLATION

3.1 Point-in-time HSTU-style user encoder

The frozen input contains separate click and purchase product-ID streams, each most-recent-first and capped at 30. It does not retain per-event timestamps or a total order across streams. We reverse each stream independently to oldest-to-newest and apply the same two-layer, four-head, 64-dimensional gated sequential-transduction stack to both. Item and stream-type embeddings are

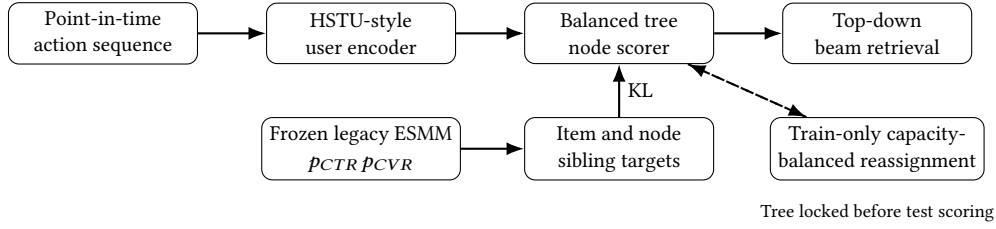


Figure 1: Registered PTD design. Solid arrows are model/data flow; dashed arrows occur only during train-side alternating optimization. The teacher is frozen.

shared, positional encoding is learned, and time encoding is disabled. Final non-padding click and purchase states are fused with a hashed-user state into z_u ; empty streams use a learned token. We use “HSTU-style” because the component adopts gated sequential transduction, not because it reproduces the scale or complete feature set of Zhai et al. [7]. A node representation and contextual features enter $s_\theta(z_u, e_n, x_{un})$.

The registered non-HSTU ablation consumes exactly the same two streams and embeddings through per-stream multiwindow DIN operators with windows (1, 2, 3, 4, 5, 5, 10). It shares the tree, scorer head, optimizer, examples, and width, but exact parameter equality is not claimed; trainable counts and measured latency are mandatory outputs. Thus RQ4 compares two fixed sequence operators without inventing unavailable temporal evidence.

3.2 Sibling teacher distributions

For a leaf-parent sibling set S_i , item-level teacher logits are

$$a_T(u, j) = \text{logit}\left(\text{clip}(q_T^{\text{purchase}}(u, j), \epsilon, 1 - \epsilon)\right), \quad j \in S_i. \quad (2)$$

For an internal child c , the teacher mass aggregates eligible descendant items from the training candidate universe,

$$m_T(u, c) = \sum_{j \in \text{desc}(c) \cap C_u} q_T^{\text{purchase}}(u, j), \quad (3)$$

and sibling logits are $a_T(u, c) = \log(m_T(u, c) + \epsilon)$. This aggregation is explicitly candidate-universe dependent; changing C_u is an experimental deviation.

The registered clipping constants are 10^{-6} for item logits and 10^{-12} for descendant mass. A child with no eligible descendant therefore receives finite, near-zero mass rather than being silently removed. Internal targets are recomputed only from the frozen training candidate universe. Multiplying every child mass in one sibling group by a common positive constant leaves its node target unchanged; the transfer is local, while the item target still reflects the teacher’s calibrated log-odds.

At temperature τ , teacher and student sibling distributions are

$$Q_T(v \mid u, S) = \frac{\exp(a_T(u, v)/\tau)}{\sum_{w \in S} \exp(a_T(u, w)/\tau)}, \quad (4)$$

$$P_\theta(v \mid u, S) = \frac{\exp(s_\theta(u, v)/\tau)}{\sum_{w \in S} \exp(s_\theta(u, w)/\tau)}. \quad (5)$$

For a purchase-positive training pair, the matched supervised path loss is

$$\mathcal{L}_{\text{tree}} = - \sum_{(u, i) \in D_{tr}: y_{ui}=1} \sum_{h=1}^H \log P_\theta(n_h \mid u, S_h(i)) \Big|_{\tau=1}. \quad (6)$$

The sibling denominator supplies path-local negatives. Every registered student uses Eq. 6; only the distillation terms, tree update, or encoder named by the ablation changes.

The complete objective is

$$\begin{aligned} \mathcal{L}(\theta; \mathcal{T}) = & \mathcal{L}_{\text{tree}} + \lambda_i \tau^2 \sum_{S_i} \text{KL}(Q_T^i \parallel P_\theta^i) \\ & + \lambda_n \tau^2 \sum_{h < S_H} \sum_{S_h} \text{KL}(Q_T^h \parallel P_\theta^h), \end{aligned} \quad (7)$$

The τ^2 factors preserve gradient scale under temperature smoothing. Item-only, node-only, combined, and no-distillation variants isolate the terms in Eq. 7. The teacher is evaluated without gradient and never updated from student or test data.

3.3 Train-only alternating optimization

The fixed tree is a deterministic binary depth-13 hierarchy over a preregistered 5,584-item catalog envelope, sorted by earliest point-in-time category and product ID. This uses test-period item identity/category but no label or teacher score; every date masks absent items before beam search. The design is therefore a disclosed transductive catalog comparison, not an inductive cold-item claim. The alternating variant follows JTM’s broad model/index co-optimization pattern but imposes a strict evidence boundary: only train-seen items may move, and only training interactions and frozen training teacher scores contribute assignment weights. Validation selects at most one of three cycles per seed; the tree-and-mask bundle is then immutable for test scoring.

For a candidate leaf ℓ with path $\pi(\ell)$, the registered train-only assignment weight is

$$W(i, \ell) = \sum_{u: (u, i) \in D_{tr}} \left(y_{ui} + q_T^{\text{purchase}}(u, i) \right) \sum_{h=1}^H \log P_\theta(\ell_h \mid u, S_h(\ell)) \Big|_{\tau=1}. \quad (8)$$

The reassignment maximizes $\sum_i W(i, A(i))$ subject to one leaf per item and the declared subtree capacities. Equation 8 lets observed purchases and frozen soft utility weight the same model-path affinity; neither test labels nor test teacher scores enter it. The executable reference supplies deterministic tie-breaking and a capacity-checking greedy solver for small examples; a production exact solver

Algorithm 1 Registered PTD training and tree lock

Require: train D_{tr} , validation D_{va} , initial tree $\mathcal{T}_0$, frozen teacher T

- 1: Verify point-in-time and candidate-universe assertions
- 2: **for** $r = 0, \dots, R$ with $R \leq 2$ **do**
- 3: Fit θ_r on D_{tr} using Eq. 7
- 4: Record validation NDCG@50 without changing T
- 5: **if** $r < R$ **then**
- 6: Compute train-only assignment weights using Eq. 8
- 7: $\mathcal{T}_{r+1} \leftarrow$ capacity-constrained reassignment
- 8: **end if**
- 9: **end for**
- 10: Select r^* using validation only; lock $(\theta_{r^*}, \mathcal{T}_{r^*})$
- 11: Score each test date once; emit manifest and evaluation JSON

may replace it only if it optimizes the same recorded weights and constraints.

This procedure is not equivalent to end-to-end differentiable tree learning. It is also not teacher-free JTM: the fixed purchase model influences local student targets and, in the alternating variant, train-side assignment weights.

3.4 Serving and training cost

Teacher inference, descendant aggregation, and capacity-constrained reassignment are offline operations. Online serving evaluates the user encoder once and then the $O(BbH)$ tree nodes above. The tree stores $O(|I|)$ nodes and one H -step code per item. Distillation increases training work because teacher scores and sibling targets must be materialized, but it does not add a teacher call or a second-stage reranker to the registered retrieval path. Latency is nevertheless measured rather than inferred from asymptotics, because batching, encoder cost, and hardware kernels can dominate node-count differences.

4 EXPERIMENTAL DESIGN

4.1 Evidence universe and splits

The verified source manifest fixes training to 2026-07-18–20, validation to 2026-07-21, and test to five dates from 2026-08-12 through 2026-08-28. The test spans 16 days. Every feature ends before its snapshot and labels occupy the next 24 hours. A metadata-only inventory pins 216 raw Parquet generations (23,761,140 rows), verifies identical split totals, and exposes the sequence, category, and feature-cutoff fields needed by PTD. A second outcome-free audit finds 3,126 train-union and 4,592 test-union items, including 2,339 absent from train and validation; it fixes the 5,584-item envelope and date masks by hash without publishing identifiers. These fields may join only on fixed row keys and cannot change eligibility or candidate counts.

Table 1 is the only empirical result currently admitted. The source manifest and evaluation JSON were independently retrieved; their SHA-256 hashes are recorded in the artifact. The table establishes a teacher/context anchor, not a PTD comparison.

Table 1: Verified legacy context on the common five-date universe. These are not PTD results.

Quantity	Verified value
Test rows	12,844,800
Purchase-positive pairs	1,136
Users with purchase	1,109
Legacy ESMM NDCG@50	0.17626
Legacy ESMM Recall@50	0.42065
Legacy ESMM AUC	0.79380

4.2 Baselines, ablations, and metrics

The registered family contains fixed-tree TDM-style supervision, item-only PTD, node-only PTD, combined PTD, train-only alternating supervision, alternating PTD, and a matched encoder ablation. We also retain an ESMM reranking oracle over the same candidate universe as a diagnostic upper anchor. TDM [9], JTM [8], beam-calibrated OTM [10], and same-level competition in DTR/TDR [2] motivate distinct baseline dimensions rather than interchangeable names.

The primary outcome is purchase NDCG@50 averaged over seeds 16630–16632. Secondary outcomes include Recall@50, NDCG@10/100, purchase AUC, click NDCG@50, diversity, candidate count, and p50/p95 latency. Date-stratified paired user bootstrap intervals use 10,000 resamples with seed 20260925. Four primary contrasts use Holm correction; the RQ2 comparator is the better single-level variant chosen on validation seed 16630 and locked before test. Every candidate’s guardrails use fixed-tree TDM: quality and diversity allow at most 5% relative degradation, and p95 latency at most 1.20 times baseline under matched hardware, single concurrency, 100 warm-up, and 1,000 measured queries. This relative threshold is not a production SLO.

4.3 Estimand and statistical procedure

For a date–user unit (d, u) , let M_{dus}^A and M_{dus}^B be purchase NDCG@50 for variants A and B under seed s . The paired seed-averaged effect is

$$\delta_{du}^{A,B} = \frac{1}{3} \sum_{s \in \{16630, 16631, 16632\}} (M_{dus}^A - M_{dus}^B), \quad \hat{\Delta}^{A,B} = \frac{1}{N} \sum_{d,u} \delta_{du}^{A,B}. \quad (9)$$

Only users with an observed purchase for the macro purchase metric enter N ; a user appearing on two dates contributes two date–user units. Each bootstrap replicate samples date–user units with replacement within each date, preserving that date’s original unit count, and then recomputes Eq. 9. The 2.5th and 97.5th percentiles form the interval. A two-sided sign probability yields the raw p -value; Holm’s step-down procedure adjusts the four RQ contrasts. Success requires a positive adjusted conclusion, a confidence-interval lower bound above zero, and every registered guardrail to pass. Secondary metrics and all slice analyses remain descriptive.

Purchase NDCG uses binary purchase relevance, logarithmic rank discount, and an ideal ranking over the same eligible list; a zero-positive query has score zero and is excluded only from the

Table 2: Preregistered prospective comparisons. Values remain pending until the evidence gate passes.

Variant	NDCG@50	Δ [95% CI]	Status
Fixed-tree TDM	–	–	pending
PTD item sibling	–	–	pending
PTD node sibling	–	–	pending
PTD item+node	–	–	pending
Alternating TDM	–	–	pending
Alternating PTD	–	–	pending
PTD baseline encoder	–	–	pending
ESMM reranking oracle	–	–	pending

registered purchase-positive macro average. Recall uses the complete eligible positive count as denominator. Category coverage and maximum category share are computed on the retrieved top 50. Latency excludes data loading and teacher computation but includes user encoding, tree traversal, and top- K materialization.

The dashes in Table 2 are intentional. One-day diagnostics cannot populate it; values require a schema-valid five-day evaluation, an immutable run manifest, and hash-linked date–user–seed records that reproduce every primary contrast.

4.4 Registered diagnostics and failure analysis

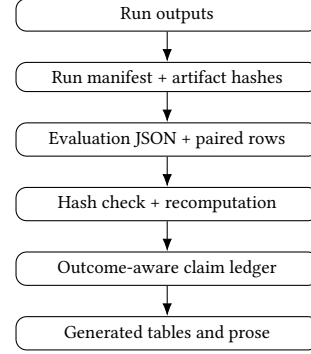
Every variant must report all seeds and dates even when it loses. The artifact records per-date metrics, cold-start cohorts, category concentration, candidates scored, and latency. Diagnostics may explain defects or failed guardrails but cannot change the primary variant, date, seed, cutoff, or tree. Missing seeds, non-finite scores, duplicate keys, universe mismatch, unlocked trees, or post-test deviations block admission. Valid null results and guardrail failures are admitted and reported but cannot satisfy positive claims; no failed slice may be deleted.

4.5 Reproducibility and claim promotion

The artifact fixes schemas for run, evaluation, and paired rows; pre-registration and amendments; a claim ledger; deterministic generators; and fail-closed validators. The checker requires all variants, seeds, dates, hashes, identities, and pre-test tree locks, then re-computes primary summaries, four bootstraps, Holm values, unit counts, and guardrail flags. Mismatch or deviation blocks admission. Admission does not imply efficacy: wording follows the verified sign, interval, adjusted p -value, and guardrails. Executable specifications and synthetic tests cover the method and statistics; SMOKE_ONLY records cannot support empirical claims. Figure 2 shows the path; a claim stays pending if an evidence edge is absent.

5 RELATED WORK

TDM introduced expressive top-down tree retrieval for recommendation and considered tree learning [9]. JTM formalized alternating optimization of index and preference model with hierarchical user representations [8]. OTM analyzed mismatch between node-wise training and beam-search inference [10], while more recent DTR/TDR work emphasizes same-level multi-class competition and sampled softmax [2]. PTD’s local softmax is related to this horizontal competition, but its target is supplied by a frozen downstream purchase model.

**Figure 2: Evidence admission and interpretation. Invalid or irreproducible bundles stop at the gate; valid negative outcomes enter the ledger without supporting positive claims.****Table 3: Claim status at manuscript generation time.**

Status	Claims
Verified	13
Preregistered	3
Pending	4

Knowledge distillation transfers behavior from a teacher into a deployable student [1]. Ranking-specific work studies listwise or top-ranked transfer [4, 6]. In text retrieval, RocketQAv2 dynamically couples a dense retriever and reranker with listwise distillation [5]. PTD differs operationally by using a frozen, purchase-oriented teacher and restricting each transfer event to siblings in a hierarchical item index. These differences do not establish novelty by themselves.

ESMM addresses click-conditioned conversion estimation with entire-space multi-task learning [3]; PTD does not modify ESMM and uses only its frozen $pCTR \times pCVR$ output. HSTU is a scalable sequential recommendation architecture [7]; here it is a user-encoder component rather than the object of the methodological claim.

6 LIMITATIONS AND THREATS TO VALIDITY

Construct validity. The ESMM score is a model-based proxy for purchase utility, not ground truth, user satisfaction, or welfare. It reflects historical exposure and can reproduce calibration, popularity, and representation biases. Internal-node mass is defined only over the frozen candidate universe and is not a catalog-wide probability. Purchase NDCG emphasizes short-horizon conversion; the click, diversity, and concentration guardrails reduce but do not eliminate this mismatch.

Internal validity. The five test dates were used previously for legacy model evaluation, so the PTD analysis is prospective only with respect to the PTD readout, not blind to legacy outcomes. The tree also uses a fixed transductive envelope of test-period item identities and categories so that later items are representable; it never uses their labels or teacher scores, and this setup does not establish inductive new-item performance. Point-in-time assertions,

Table 4: Positioning against primary adjacent work. PTD statements describe the registered design, not empirical superiority.

Work	Prior mechanism	Registered PTD distinction
TDM [9]	tree node / item leaf; teacher: none; tree: optional learned tree.	PTD retains coarse-to-fine beam retrieval but changes supervision to purchase-aware sibling distributions.
JTM [8]	tree node / item leaf; teacher: none; tree: alternating joint optimization.	PTD adopts only train-side alternating structure learning and adds a frozen external purchase teacher.
OTM [10]	beam-visited nodes; teacher: estimated optimal pseudo-target; tree: not PTD’s focus.	OTM targets beam-search calibration; PTD targets transfer of a downstream purchase model.
DTR/TDR [2]	same-level competing nodes; teacher: none; tree: learned tree / forest.	Both use horizontal competition; PTD’s soft targets come from fixed purchase scores over sibling groups.
ESMM [3]	impression/item score; teacher: not applicable; tree: none.	PTD freezes ESMM pCTR x pCVR as teacher rather than proposing a new conversion model.
Ranking KD / RankDistil [4]	ranked list; teacher: high-capacity ranker; tree: none.	PTD specializes listwise transfer to local tree siblings and evaluates candidate retrieval.
RocketQAv2 [5]	passages; teacher: dynamic reranker/retriever; tree: none.	Both connect ranking and retrieval; PTD uses a frozen purchase teacher and a hierarchical item index.
HSTU [7]	sequential recommendation tokens; teacher: none; tree: none.	PTD uses an HSTU-style encoder as a component; it does not claim the HSTU architecture.

fixed row keys, a frozen teacher, validation-only selection, immutable hashes, date masks, and a pre-test tree lock address identifiable leakage paths. They cannot rule out defects in upstream logging or unrecorded feature generation; a hash proves identity, not semantic correctness.

Statistical conclusion validity. Purchases are sparse and five date strata limit temporal replication. Pairing shared requests and seeds reduces variance but does not add dates. Holm correction covers only the four registered primary contrasts. Slice analyses, secondary outcomes, and the teacher oracle are descriptive; absence of significance is not evidence of equivalence.

External validity. One catalog, one historical teacher, a fixed tree family, and one beam/latency protocol limit generalization. Offline NDCG establishes neither online causal lift nor transfer to other catalogs and hardware. Until the five-day artifact is complete, this draft cannot support acceptance-level empirical conclusions; a positive result would still require a separately governed online experiment.

7 ETHICS AND SOCIETAL CONSIDERATIONS

Purchase optimization can over-emphasize high-propensity users or products, amplify exposure inequality, and encourage consumption. We record concentration and cold-start cohorts and forbid removal of failing slices. Private paired evidence uses access-controlled pseudonyms; the public artifact contains no row behavior. Review artifacts must be anonymous and policy compliant. Deployment requires separate safety and rollback review; an offline decision never changes production.

8 CONCLUSION

We specified Purchase-aware Tree Distillation as a testable integration of a frozen ESMM purchase teacher, sibling-level tree supervision, an HSTU-style sequence encoder, and train-only alternating tree optimization. The current contribution is methodological and procedural: equations, algorithm, preregistered contrasts,

and a machine-checkable evidence boundary. **[Pending: Five-day PTD effect sizes, confidence intervals, latency measurements, and ablation conclusions await a valid artifact.]**

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